Cardiovascular Disease Forecast using Machine Learning Paradigms

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***Abstract*— In this recent era, Cardiovascular disease (CVD) propagation rate has been intensifying the cause of death worldwide among the non-communicable disease. In particular the south asian countries have a tremendous risk of cardiovascular disease at an early age than any other ethnic group. Most often it's challenging for medical practitioners to predict cardiovascular disease as it requires experience and knowledge which is a complex task to accomplish. This health industry has enormous amounts of data which is useful for making effective conclusions using their hidden information. So, using appropriate results and making effective decisions on data, some superior data analysis techniques are used, for example Naive Bayes, Decision Tree. By using some properties like (age, gender, bp, stress, etc) it can be predicted the chances of cardiovascular disease. In this study, we collected 301 sample data with 12 clinical attributes. Logistic regression, Decision tree, SVM, and Naive bayes classification algorithms have been applied to predict heart disease. In this case, logistic regression provided 86.25% accuracy. However, we also compared the UCI dataset based results with our model.**

Keywords*— *Classification Algorithm, Heart Diseases, Decision Tree, SVM, Logistic Regression, Naive Bayes, UCI dataset.

I. INTRODUCTION

Cardiovascular disease is one of the leading causes of death throughout the world. Many people die from cardiovascular diseases than from any other causes. Cardiovascular diseases are more accountable than from any other causes to lay down one's life. It accounts for nearly one in every three deaths worldwide annually according to The World Health Organization report. If the heart stops functioning normally, our body's other organs will stop their working process. Mortality rate increases on account of cardiovascular disease in different countries including Bangladesh. In 2017, The World Health Organization figured out the deaths from heart disease in Bangladesh entered 14.31 percent of total deaths, and every year cardiovascular disease kills 17.9 million people, 31 percent of global deaths [1].

In the medical domain, data mining can be used to extract information from hidden patterns of dataset. In present we receive any medical data in a distributed way as a paper document. This data orientation needs an embodied structure. Pre-process data is needed to apply machine learning algorithms to gain a more accurate result. By using data mining techniques, this extractive data will help to predict the medical diagnosis. Future prediction based approach also will help the doctor's to take the right steps to treat the patient in time with the help of previous patterns of the dataset. Techniques of the data mining and model of the prediction are responsible for proper prognosis of the disease [2].

Cardiovascular disease rate increases gradually in related to people's driven life-style as like habits of smoking, high fat ingestion, lakings of physical mobility. As the heart is a pump to circulate the blood across the whole body. According to The United States National Institutes of Health (NIH) reports heart rate varies from person to person on the basis of different parameters on average 60-100 times in one minute [3].

This study is for finding the appropriate model to predict cardiovascular disease. Heart disease is one of the major causes of death in our country also. It is crucial to make people aware about the risk factor of cardiovascular diseases.

In the following section we first discussed a few related works then talked about our proposed approach to find the risk factor as well as a better model for heart disease prediction. After that, in section III we discussed the result, and next conclusion with some notes for future work in section IV.

II. RELATED WORK

This section is for presenting the research demand on this topic and some works that must highlight our study. We found that a lot of research was focused on cardiovascular disease. We were keen to know the risk factors for predicting heart disease.

Nabaouia Louridi, Meryem Amar and Bouabid El Ouahidi used different machine learning algorithms to identify the Cardiovascular Disease (CVD) where they finally proposed a SVM with a linear kernel approach. In this approach they used 13 features and found an accuracy of 86.8% [4]. In 2019 N. Satish Chandra Reddy, Song Shue Nee, Lim Zhi Min & Chew Xin Ying stated that Random Forest can be used as classification algorithm to train the system for identifying CVD with 90% to 95% accuracy. They used 14 features [5].

Using the Dataset of UCI library in 2018 Aditi Gavhane, Gouthami Kokkula, Isha Pandya & Prof. Kailas Devadkar conducted a study to predict the heart disease of a human using 13 features among 76. In this study they used MLP and got an average of 0.91 precision [6].

In this work, Sonakshi Harjail & Sunil Kumar Khatri have used Correlation-based feature selection and Multilayer Perceptron classifier to propose a new model. 297 input was tested to find an 89.2% accuracy with the proposed model [7].

In 2019 Senthilkumar Mohan, Chandrasegar Thirumalai And Gautam Srivastava proposed a Hybrid Random Forest with a linear model with 88.7% accuracy to predict heart disease. In this study they consider UCI machine learning repository to collect the dataset. They used multi-class variable and binary classification for data pre-processing [8].

Uma N Dulhare in 2018, narrates a methodology to improve the performance of Bayesian classifier to predict heart disease. Here, the author used Statlog (heart) data set UCI where there were 14 attributes with 1 class label and 270 instances with no missing values. He obtained 87.91% accuracy for Particle swarm optimization (PSO) with Naive Bayes classifier [9].

So, it is obvious that we need a proper solution to prevent heart disease as soon as possible. This is because all over the world researchers are very keen to work with cardiovascular disease prediction as we are able to know the risk factors of this disease.

III. Proposed Model

In this study we address the issue for predicting heart disease. We collected 301 data against 13 attributes [10]. This study proposed an approach which classified the risk of having cardiovascular disease. In this proposed system, we used Logistic regression, Naive Bayes, SVM, and Decision Tree classification algorithm for getting better accuracy in the case of predict heart disease. Finally, analysing the obtained results with the help of Comparing Models through Confusion Matrix.

A entire of 301 sampless with 13 attributes were obtained as we mentioned. After that, whole samples were divided equally into two sets: training data (80%) and testing data (20%). To avoid bias, the samples for each set were nominated randomly. Presented Fig. 1 to illustrate the whole working process. We started our study with feature selection as we considered UCI dataset (Heart) to compare our study. After feature selection we collwcted all sample data . At the end of the study, we applied four selected classifier algorithms to predict the risk level of cardiovascular disease. Among four classifier we got best result from logistic regression. Finally, our model is ready to predict heart disease.

There are some machine learning algorithms from the area of statistics. Also known as the go-to technique for classification problems in machine learning. Logistic Regression is a little bit similar because both have the goal of estimating the values for the parameters or coefficients. After training a model we find out the relation between training and testing data. We got 86.25% accurate result for logistic regression.

Naive Bayes is another classification algorithm which uses the Bayes theorem with independent assumptions between features. One dimensional Naive Bayes classifier computes the ratio of the log probabilities of the features belonging in all the classes. Naive Bayes does not consider the correlation between attributes. Naive Bayes is a very scalable classifier but it can create a bias towards one or more attributes. In our approach we found 73.77% accuracy after applying naive bayes.

In the large amount of data environment Support vector Machine (SVM) provides classification learning model. Linearly (e.g., straight line or hyperplane) data domain dividation is called Linear SVM. And data domain transformation to a space entitled the space of feature and to separate the classes, data domain can be divided linearly is named non linear support vector machine [11]. In the case of classify our dataset we got 83.61% accuracy for SVM.

Finally, our last selected algorithm was Decision tree. Decision tree is one of the most popular classification and decision making algorithms. Different types of decision tree are available. Here, we used classification decision tree to classify the datase. However we did not get better result (75.41%) for this algorithm. Arundhati Navada et al. in 2011 presented a paper for describing the basic idea of decision tree. In this paper they talked about different types of decision tree with equation and explanation [12].

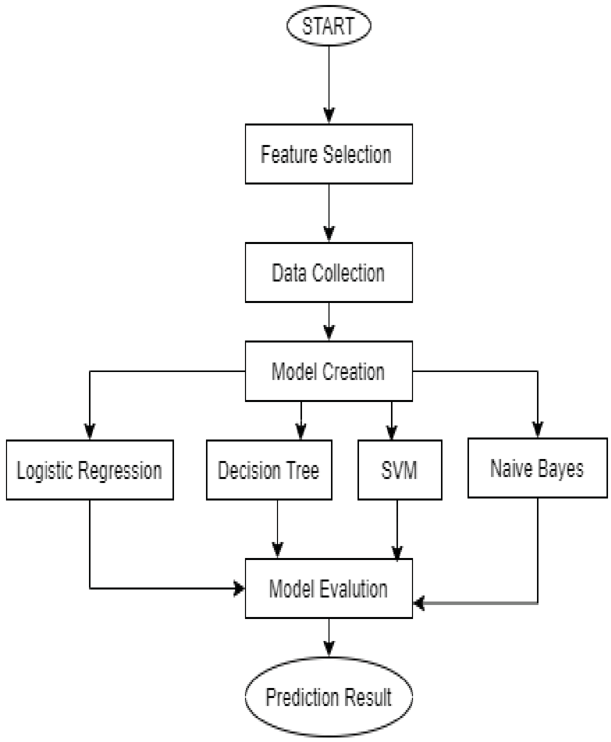

Fig. 1. Flowchart of our proposed Model

Now, we are going to describe our datset. We considered UCI repository as a standard clinical attributes to select features for our study as we said. After that we fixed our attributes list with class label to collect the data from a google form. Table 1 for presenting the attributes list of our dataset. Here, we described all 12 attributes with proper measurement values.

TABLE 1. DESCRIPTION OF ATTRIBUTES

Attribute Description	Encodings
Gender	Male (1), Female (0)
Age(Years)	20-34 (-2), 35-50 (-1), 51-60 (0), 61-79 (1), >79 (2)
Blood Cholesterol	Below 200 mg/dL - Low (-1) 200-239 mg/dL - Normal (0) 240 mg/dL and above - High (1)
Blood Pressure	Below 120 mm Hg- Low (-1) 120 to 139 mm Hg- Normal (0) Above 139 mm Hg- High (-1)
Hereditary	Yes (1) No (0)
Smoking	Yes (1) No (0)
Alcohol Intake	Yes (1) No (0)
Physical Activities	Low (-1), Normal (0), High (-1)
Diabetes	Yes (1) No (0)
Diet	Poor (-1), Normal (0), Good (1)
Obesity	Yes (1) No (0)
Stress	Yes (1) No (0)

Now, we are going to explore our dataset. Number of affected people according to gender and comparison of age for blood cholostore is illustrated in Fig. 2 and 3 respectively for depicting the dataset.

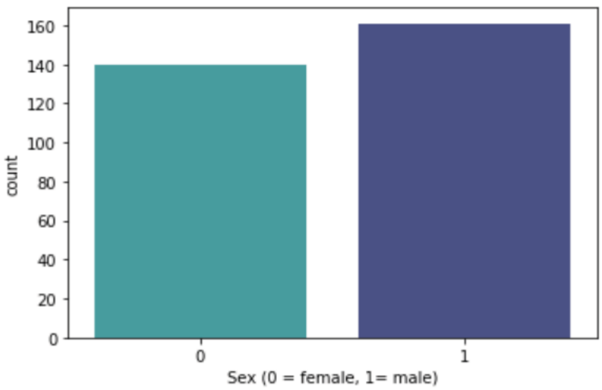

Fig. 2. Explore Gender Dataset

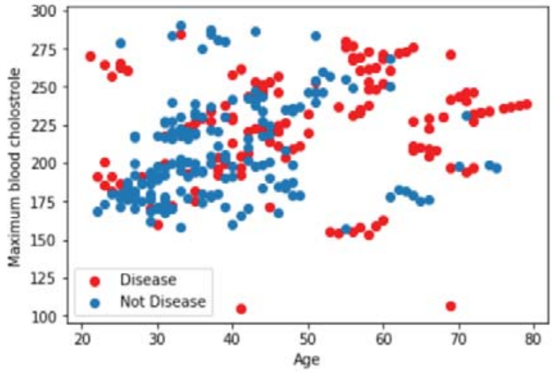

Fig. 3. Explore Blood Cholesterol Dataset

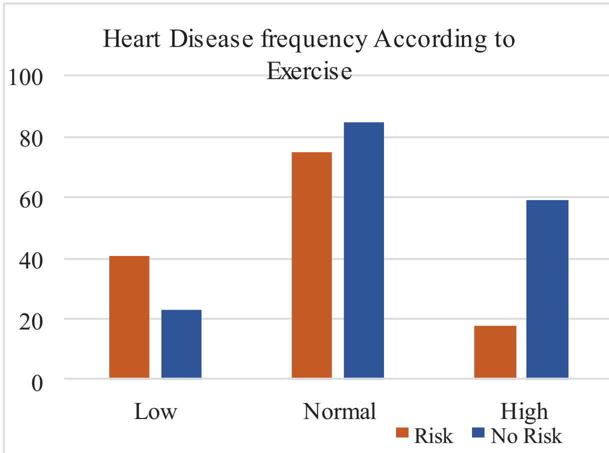

Fig. 4. Explore Class label for exercise.

Exercise or physical activity is one of the most effective factor to predict heart disease. In Fig. 4, we presenting class label according to daily exercise. It is obvious from bar chart thet exercise is helpful to reduce risk of heart disease.

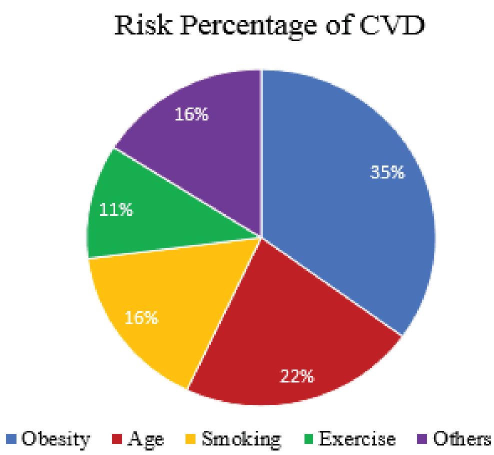

Fig. 5: Risk Factor of CVD

We demonstrated the risk factor of Cardiovaascular Disease (CVD) with the help of Fig 5. Here, we got that obesity is one of the crucial risk factors for CVD.

IV. RESULT ANALYSIS

Our prediction approach was developed with 13 attributes with class labels. Table 2 for present the result of our proposed approach. Here, it is obvious that logistic regression gave better results among four selected models. In our strudy we applied all those classifier on our own collected data. Elsewhere we found more accurate those who worked with dataset (Heart) from UCI repository.

TABLE 2. COMPARISON THE ACCURACY OF VARIOUS ALGORITHMS

Classification Algorithm	**Accuracy**
Logistic Regression	86.25%
Support Vector Classifier	83.61%
Decision Trees	75.41%
Naïve Bayes	73.77%

A. Comparative Study

In this study we also ovserved UCI Heart disease dataset and foun many reseach works on this dataset. Senthilkumar Mohan et al. worked with UCI dataset and they also applied those four classification algorithms. Table 3 for compared our result with UCI dataset based study.

TABLE 3. UCI DATASET BASED RESULT COMPARISION

Classification Algorithm	Accuracy
Logistic Regression	82.9%
Support Vector Classifier	86.1%
Decision Trees	85%
Naïve Bayes	75.8%

Here, we can find that they got less accuracy for Naive bayes and best result for SVM. Howwever, we got best result for logistic regression. Now Table 4 to depict the result of Logistic regression classifier.

TABLE 4. LOGISTIC REGRESSION RESULT ANALYSIS

Class Label	Precision	Recall	F1-score	Support
0	0.82	0.97	0.89	34
1	0.95	0.74	0.83	27
Avg	0.89	0.86	0.86	61

V. CONCLUSION

Heart disease is one of the major deaths anywhere in the world. We addressed this issue in addition proposed an approach to predict the risk factor as we can prevent this disease as early as possible. In this study we got logistic regression provided better results against 12 attributes. We also compared UCI dataset and found the factor to predict heart disease. Many research works showed more accurate results though they used UCI repository. However, we considered our own collected data.

In future we have a desire to predict this disease with more accurately and for this case we will have to collect more data as much as possible. Concurrently, we need accurate clinical data to predict any types of diseases. Everyone needs good health to live a beautiful life otherwise social development will be stuck.

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